

Virtual Lab 02

Scientific Reasoning and Hazardous Materials Investigation

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Table of contents

Purpose	1
Skills	2
Course Learning Outcomes	2
Instructions	2
Part 1: Scientific Reasoning	2
Part 2: Data Analysis and Visualization	3
Part 1: Incident Investigation	3
Historical Scenario: February 22–24, 1978	3
Scientific Reasoning Questions	4
Question 1: What has happened?	4
Question 2: What will or could happen?	4
Question 3: What can you change?	4
Part 1: Submission Requirements	4
Part 2: Data Analysis and Visualization	5

Purpose

The purpose of this laboratory is to develop your ability to apply scientific reasoning to a real-world hazardous materials incident.

Hazardous materials management requires more than identifying a chemical or memorizing its properties. Safety and health professionals must use available evidence, recognize uncertainty, anticipate how hazardous materials may behave, and make informed decisions to protect workers, emergency responders, and surrounding communities.

In this laboratory, you will investigate a historical railroad incident involving liquefied petroleum gas (LPG).

You will practice answering three fundamental questions:

1. **What has happened?**
2. **What will or could happen?**
3. **What can you change?**

The objective is not simply to identify the correct answer. It is to demonstrate how you arrive at your conclusions using scientific reasoning.

Skills

By completing this laboratory, you will practice the following skills:

1. **Scientific observation:** Identify important facts, environmental conditions, and changes in an incident.
2. **Critical thinking:** Distinguish between observations, assumptions, and conclusions.
3. **Scientific reasoning:** Apply principles of chemistry and physics to understand hazardous materials behavior.
4. **Hazard anticipation:** Predict potential outcomes using available evidence.
5. **Problem-solving:** Identify actions that could prevent or reduce the consequences of a hazardous materials incident.
6. **Technical communication:** Explain your reasoning clearly and support conclusions using scientific principles.

Course Learning Outcomes

Upon completion of this laboratory, you should be able to:

1. Explain how physical and chemical properties influence the behavior of hazardous materials.
2. Apply scientific reasoning to investigate a hazardous materials incident.
3. Identify conditions that may contribute to a hazardous materials release.
4. Predict potential consequences of hazardous materials under changing environmental conditions.
5. Recommend preventive or corrective actions based on available scientific evidence.
6. Communicate technical conclusions using clear, evidence-based reasoning.

Instructions

Virtual Lab 02 consists of two parts.

Part 1: Scientific Reasoning

You will review a historical railroad incident involving liquefied petroleum gas (LPG).

Using the information provided, you will answer three questions designed to develop your scientific reasoning abilities.

Your responses should demonstrate how you think through the problem, not simply state what happened.

You are encouraged to use your lecture materials and credible scientific resources.

You do not need to use RStudio, Excel, or any other data analysis software for Part 1.

Part 2: Data Analysis and Visualization

In Part 2, you will receive a dataset related to the incident.

You will use RStudio, Microsoft Excel, or another software program of your choice to create visualizations and interpret the data.

You will then revisit your initial scientific reasoning and evaluate your conclusions using quantitative evidence.

Part 2 instructions and the dataset will be provided separately.

Part 1: Incident Investigation

Historical Scenario: February 22–24, 1978

On February 22, 1978, at approximately 10:30 PM, 24 freight cars derailed in a downtown area.

Weather conditions included light snow on the ground and temperatures in the mid-20s degrees Fahrenheit.

The freight train was traveling at the posted speed limit of 35 miles per hour.

The rear brakeman, positioned toward the front of the train, noticed an automatic emergency brake application. He observed that the train had separated and that cars 17 through 39 had derailed.

Among the derailed freight cars were two tank cars containing liquefied petroleum gas (LPG):

- UTLX-83013
- UTLX-81467

Upon initial inspection, neither tank car displayed evidence of leakage, although both had sustained substantial damage.

It was decided that the tank cars would be repositioned so their contents could be transferred into highway tankers.

Several freight cars obstructing access to the LPG tank cars were moved.

A cable sling was attached to one end of an LPG tank car, with the opposite end used as a pivot point. A wooden support was placed beneath the end where the cable was attached.

The other cars were repositioned using similar methods.

The tank car relocations were completed by approximately 2:15 PM on February 23.

Gas monitoring conducted during the recovery operations showed no indication of leakage.

Recovery crews continued removing damaged freight cars and wreckage until approximately 8:00 PM on February 23.

Because of the extent of its damage, the LPG tank car at the southern end of the incident was selected for unloading first.

The second LPG tank car was scheduled for unloading the following day, February 24.

By approximately 1:00 PM on February 24, a highway tanker arrived to facilitate the transfer.

At approximately 2:53 PM, the northern LPG tank car ruptured, releasing LPG into the surrounding environment.

The released gas ignited shortly afterward from an undetermined ignition source.

By the time of the rupture, outdoor temperatures had increased into the 50s degrees Fahrenheit, and skies were clear.

Scientific Reasoning Questions

Using the information provided and the scientific principles introduced in this course, answer the following questions.

Question 1: What has happened?

Identify the important observations and conditions associated with the incident.

Explain which information you believe is relevant to understanding the situation.

Distinguish between what is known and what you may need to investigate further.

Question 2: What will or could happen?

Using your understanding of chemistry, physics, and the behavior of matter, explain what you believe could happen as conditions change.

Develop a scientifically reasonable prediction based on the available evidence.

Explain how you arrived at your prediction.

Question 3: What can you change?

Imagine that you are the safety and health professional responsible for evaluating the incident.

What actions would you recommend to reduce the likelihood or severity of a hazardous materials incident?

Explain the scientific reasoning behind your recommendations.

Identify any additional information you would need before making a final decision.

Part 1: Submission Requirements

Submit a document containing your responses to all three scientific reasoning questions.

Your responses should:

- Demonstrate your thought process.
- Use relevant scientific principles.

- Distinguish observations from assumptions.
- Explain the reasoning supporting your conclusions.
- Identify uncertainties when appropriate.

The emphasis is on scientific reasoning, not simply identifying a correct answer.

Part 2: Data Analysis and Visualization

Part 2 will be released separately.

In this portion of the laboratory, you will examine quantitative data associated with the physical behavior of LPG.

You will create visualizations using a software program of your choice and use the resulting evidence to evaluate your initial predictions.

Further instructions and the dataset will be provided when Part 2 is released.

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